Solar Filament Instance Segmentation under Annotator Disagreement: Soft Multi-Annotator Targets, DINOv3 Members and PQ-Additive Instance Selection

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***Abstract*—We segment solar filaments in 2048×2048 GONG Hα full-disk images. The metric is Panoptic Quality (PQ), computed separately against every annotator of an image. Two annotators of the same image agree at only PQ $= 0.340$, so annotator disagreement, not pixel accuracy, limits every honest system. We train U-Net and DINOv3 networks on soft consensus targets. We show that PQ is exactly additive over disjoint predicted instances, and we use this to tune post-processing and to train instance scorers with nested cross-validation. Our final system (ResNet-34, EfficientNetV2-S and DINOv3 ConvNeXt-S members, a hypothesis pool with a learned fragment linker and a stacked scorer) reaches a nested out-of-fold PQ of 0.4477 on the 707 training images (public leaderboard 0.39). We report ceilings, an error breakdown and an analysis of the leaderboard gap.**

I. Data

The training set (MAGFiLO [1]) has 707 images with 1,154 annotation sets and 8,199 filament polygons. 411 images have one annotator, 145 have two and 151 have three. Filaments are thin and elongated (median area 1,228 px, 5th percentile 317 px). The 180 test images come from the same six GONG sites and the same years (2011–2022) in similar proportions. The median gap to the nearest training image is $\sim$37 h, as between training images, so we treat the test images as a random sample of the same observation pool. Each annotation set is identified by `<annotator>-<image>`. The 37 annotator codes (a 4-digit group and a 2-digit slot) have clearly different styles: from 3.4 to 10.5 polygons per image and a median filament area from 640 to 2,100 px.

II. Metric analysis

For every annotation set, all predicted filaments of the image are compared with its ground truth (GT). Pairs with IoU > 0.5 are true positives (TP) and contribute their IoU. Counts are pooled over all sets: PQ $= \sum \mathrm{IoU}_{TP}/(TP + \frac{1}{2}FP + \frac{1}{2}FN)$ [2]. Our implementation reproduces the organisers' notebook exactly.

Ceilings. Table I shows that even an oracle that knows every label cannot exceed 0.666 with one set of masks on multi-annotator images, because the annotators contradict each other. Our system lies above human consensus because it predicts the average annotator.

TABLE I
Human agreement, ceilings and our system (organisers' metric).

reference	PQ	SQ	RQ
annotator vs. annotator (same image)	0.340	0.634	0.536
consensus of two vs. held-out third (151 images)	0.333	0.632	0.528
oracle with all labels known (296 multi-set images)	0.666	0.825	0.807
ours (nested out-of-fold, 707 images)	**0.448**	0.683	0.655

PQ is additive over disjoint instances. If predicted masks do not overlap, a GT filament can be hit by at most one prediction. Hence for any subset S of predictions

$$\mathrm{PQ}(S) = \frac{\sum_{i \in S} a_i}{\frac{1}{2}N_{gt} + \sum_{i \in S} d_i}, \tag{1}$$

where a_i is the sum of instance i's TP IoUs over the m_i annotation sets of its image and $d_i = m_i/2$. Adding instance i raises PQ if and only if its expected per-annotator TP IoU $u_i = a_i/m_i$ exceeds $\mathrm{PQ}/2$ (≈ 0.22). After extracting instances once, every threshold and filter search is an exact sum. The metric counts every IoU > 0.5 pair, so duplicate masks inflate it: submitting each mask twice would raise our out-of-fold PQ from 0.4385 to 0.4831. **We submit strictly disjoint masks.**

III. Method

Targets. Polygons are rasterised exactly like the GT (pycocotools). The soft target of a pixel is the fraction of annotators covering it. We also derive a soft skeleton, a soft core and one instance map per annotation set. Only the `segmentation` field is read.

Networks (Table II). All members see the full disk as one channel. The v3 members share a decoder of GroupNorm/GELU/scSE [7] blocks with a light stride-1/2 detail stem, skeleton and core heads, and deep supervision. M4 uses a DPT decoder [4] on DINOv3 ViT-B/16 [3].

Losses. The first two members use BCE + soft Dice against the soft target. M3–M5 use BCE + the Jaccard metric loss JML1 [5], which is minimised at $P = y$ for soft labels. They add a blob loss [6] over the instances of one sampled annotation set, so that every filament counts equally as in PQ,

TABLE II
ENSEMBLE MEMBERS. OOF PQ OF EACH MEMBER ALONE, AT THE DEFAULT EXTRACTION THRESHOLD 0.6 (TUNED VALUE IN BRACKETS).

member	encoder (pretraining)	px	OOF PQ
r34_2048	ResNet-34 (ImageNet), U-Net	2048	0.4331
effv2s_1536	EfficientNetV2-S (ImageNet), U-Net	1536	0.4355
M3	ResNet-34 (ImageNet), U-Net	1536	0.4131 (0.4303)
M4	DINOv3 ViT-B/16, DPT	1536	0.4268 (0.4410)
M5	DINOv3 ConvNeXt-S, U-Net	1536	0.4303

TABLE III
NESTED OUT-OF-FOLD PQ (707 IMAGES) AND PUBLIC LEADERBOARD.

system	PQ	SQ	LB
r34 @1024, fixed threshold, no TTA (fold 0)	0.380		
+ TTA, tuned thresholds and filters	0.4205		
+ rescorer	0.4272		
v1: 3-member ensemble + rescorer	0.4381	0.677	0.38
v2: r34_2048 + effv2s_1536 + extent selector	0.4438	0.680	0.38
v2 selector, + M3	0.4449	0.682	0.38
v2 selector, + M5	0.4455		
final: Stage C, r34_2048 + effv2s_1536 + M5	**0.4477**	0.683	0.39

plus skeleton/core head losses and deep supervision. A 5-fold proxy ablation (ResNet-34, 1024 px) with a paired bootstrap gate chose this loss: BCE + soft Dice 0.3861, BCE + JML1 0.3864 (not significant), full v3 loss **0.3956** ($+0.0092$, 90 % CI $[+0.006, +0.012]$).

Training. AdamW, warm-up and cosine decay, batch 8, bf16, weight EMA (0.998), 60–80 epochs. DINOv3 encoders use a lower encoder learning rate with layer-wise decay. Images are sampled in proportion to their number of annotation sets, as the metric weights them. Augmentation runs on the GPU: rotation, flips, scale, shift, gamma, contrast, blur and noise.

Validation. 5 image-level folds, so all annotation sets of an image share a fold. Every learned post-processing stage is *nested*: for fold k it is trained on the other folds' out-of-fold predictions.

Instances. Maps are averaged over 8 dihedral TTA views and the 5 fold models of each member, then over the members. Seeds are the components of $p > t$. Seeds within $\sim$16 px are grouped into one filament, because annotators draw one polygon around a broken filament ($+0.008$ PQ). The extent is grown by watershed, with holes filled. The v3-loss members produce softer maps (mean foreground probability 0.80 vs. 0.90), so their best threshold is 0.4 rather than 0.6 (Table II).

Selection. (i) A LightGBM [8] rescorer regresses u_i from 26 instance features and keeps $u_i > t$; the tuned t matches the theoretical PQ$/2$. (ii) An extent selector picks one of four threshold levels per filament. (iii) Stage C, used in the final system, builds a pool of hypotheses: threshold families and a learned fragment linker at two link thresholds. A stacked LightGBM scorer then ranks the candidates, followed by an overlap-aware greedy selection.

IV. RESULTS

Table III shows the progression. Per-fold validation PQ (single model, no TTA): M3 0.367–0.392, M4 0.385–0.406, M5 0.371–0.412, so the DINOv3 networks are the strongest single members. The selection stages add little: rescorer $+0.007$, extent selector $+0.0004$ to $+0.003$, Stage C $+0.0022$ over the selector on the final members. Their oracles show 3–8 points of headroom, but which extent a given annotator draws is largely unpredictable from the image.

Rubric metrics (Fig. 1). 75.7 % of overlapping pairs have IoU > 0.5. The mean/median IoU is 0.595/0.646 and the mean/median Dice 0.726/0.785. Fragmentation (GT filaments overlapped by 2+ predictions) is 3.0 % and over-merging (predictions overlapping 2+ GT filaments) 2.9 %.

Where PQ is lost. 56 % of the lost PQ comes from the IoU deficit of true positives (boundary disagreement), 15 % from missed filaments, 14 % from wrong extent (IoU ≤ 0.5), 6 % from spurious predictions and 10 % from fragments and merges. Small filaments are the weakest group (< 500 px: PQ 0.29, recall 0.36), followed by the limb ($r/R > 0.8$: PQ 0.37). PQ is uniform across sites (0.43–0.45) and across the number of annotators per image (0.44/0.46/0.44). It is *not* uniform across annotator groups (0.40–0.55).

Leaderboard gap. Our submissions score 0.38–0.39 publicly, $\sim$0.06 below out-of-fold, and other teams report the same gap. It is not chance: random 90-image subsets of our out-of-fold predictions have SD 0.012 and fall to ≤ 0.385 with probability < 0.0003. It is not mask width: the same masks eroded/dilated by 1 px score 0.37/0.35. It is not calibration: test maps differ from out-of-fold maps by $\sim$1 % in foreground fraction. The most likely cause is an annotator shift, since our PQ varies by 0.15 between annotator groups. We tuned nothing on the leaderboard.

What did not work. Keeping only the largest component (undoes the $+0.008$ of fragment grouping), per-instance closing (-0.003 to -0.016), multi-scale TTA ($+0.0006$) and JML1 alone ($+0.0003$) were not adopted. Blend weights searched at the v1 threshold 0.6 excluded M4, the best single member; tuning the threshold per blend fixes this. Not run because of the compute budget: semi-supervised training on unlabelled GONG frames and ConvNeXt-L/ViT-L members.

V. COMPLIANCE, EFFICIENCY AND REPRODUCIBILITY

At inference the model sees only the test image. Of the GT fields, only `segmentation` is used. The public MAGFiLO v1.0 Kaggle dataset contains 139 of the 180 test images with labels; it was *never downloaded or used*. No other external data were used, and masks are disjoint. Pretrained weights: ImageNet and DINOv3 (DINOv3 License; our fine-tuned DINOv3 checkpoints are released with the licence text).

On one AMD Instinct MI300X, training 5 folds $\times$ 60 epochs takes 1.3 h (M3, 3 folds in parallel), 5.1 h (M5) and 6.0 h (M4). Inference with 8 TTA views takes 0.46–0.72 s per image and fold model. Post-processing and selection for 180 test images take 7 s on 12 CPU workers.

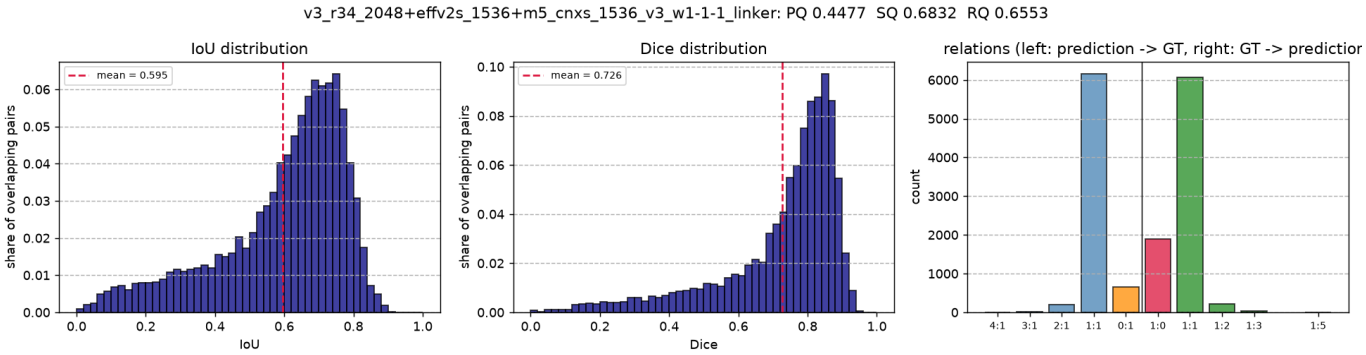


Fig. 1. Final system, out-of-fold: IoU and Dice distributions of overlapping prediction–GT pairs, and the relations between predictions and GT filaments (1:1 dominates; 1:n and n:1 are rare).

Code, configurations, logs and the reproduction notebook are at https://github.com/umerkang66/solar-filament-segmentation. The final file is reproduced from the released weights by submission_package_v3/notebook/reproduce_v3.ipynb. Training: `SETUP.md`, `scripts/run_all_v3.sh`; decisions: `DECISIONS.md`.

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This work uses DINOv3 (Meta) pretrained weights under the DINOv3 License, and GONG Hα data from the NSO Integrated Synoptic Program (MAGFiLO dataset).

References

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